

AI and the Distribution of Prosperity: A Unified Framework for How Technology Shapes Inequality

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Full paper in progress

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Motivation

- Previous technological waves (computerization, the internet, and industrial robots) were accompanied by sharp increases in inequality.
 - Incomes at the top of the U.S. distribution grew rapidly, while incomes for the bottom half mostly stagnated or even declined (Piketty et al., 2018; Moll et al., 2022).
 - As a result, the top 1% increased its share of national income from 12% in 1978 to 20% in 2018, while the bottom 50% experienced the opposite trend (Saez and Zucman, 2019).
- How AI's gains are distributed could have profound economic, social, and political consequences for decades to come.

Through which mechanisms can AI affect income inequality?

What determines their direction and magnitude?

Setup: A Standard Decomposition of Inequality

The top group's share of total income:

$$S_{\text{top}} \equiv (1 - s_K) c_L + s_K c_K$$

Any effect of AI on inequality must run through three channels:*

$$\Delta S_{\text{top}} = \underbrace{\Delta c_L (1 - s_K)}_{\text{wage concentration}} + \underbrace{\Delta c_K s_K}_{\text{capital concentration}} + \underbrace{\Delta s_K (c_K^{\text{postAI}} - c_L^{\text{postAI}})}_{\text{shift toward capital}}$$

*Or through changes in relative prices.

This Paper: A Unified and Accessible Framework

1 Direct Effects on Production

Automation · Augmentation · New-task creation
Entry barriers · Capital deepening

KEY QUESTIONS

- What can AI do, and how will firms use it?
- Which workers and tasks are affected?
- How much capital is used, and who owns it?



2 Who Captures the Initial Productivity Gains?

Workers (wages) · Adopting firms (profits)
AI suppliers (profits) · Consumers (lower prices)

KEY QUESTIONS

- How competitive are product and AI markets?
- How much bargaining power do workers have?
- How much of the gain passes through to prices?
- Who gains within each group?

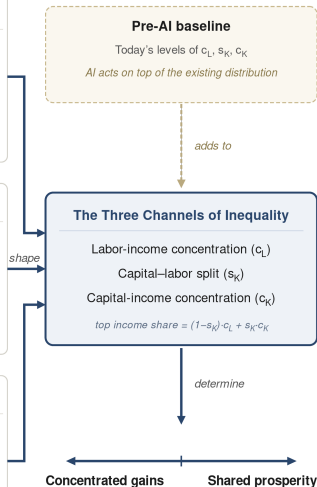


3 How the Gains Spread Through the Economy

Spending and saving create new demand
across sectors and for new kinds of work

KEY QUESTIONS

- Who initially receives the gains?
- What do they consume, and how much do they save?
- Which sectors and workers meet the new demand?



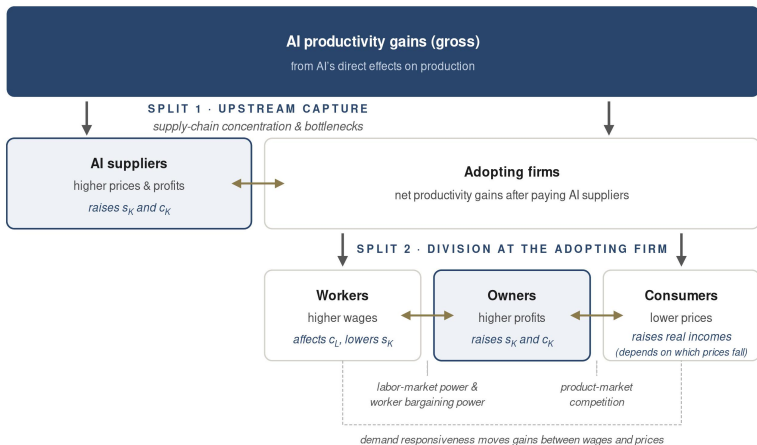
Stage 1: Direct Effects on Production

Parallels **Acemoglu et al. (2026)**:

Mechanism	Description	The key distributional questions we discuss
Automation (§3.1.1)	AI performs existing tasks end-to-end with little human intervention, substituting capital for labor.	1. How far does automation extend? (s_K) 2. Which workers are directly affected? (c_L) 3. Where do displaced workers move? (c_L) 4. Who owns the capital that gains? (c_K) 5. Who captures the productivity gains? (c_L, s_K, c_K)
Augmentation (§3.1.2)	AI raises the productivity of workers in tasks they continue to perform.	1. Which workers experience the largest productivity gains, within and across occupations? (c_L) 2. Who captures the productivity gains? (c_L, s_K, c_K)
New-task creation (§3.1.3)	AI creates entirely new tasks and occupations.	1. How much new work is created? (s_K) 2. Who gets the new work? (c_L, s_K, c_K)
Changes in entry barriers (§3.1.4)	AI changes the expertise needed to perform tasks or enter occupations, altering occupational wage premiums.	1. Where are barriers lowered or raised? (c_L) 2. Who can access and use AI effectively? (c_L) 3. Do institutions allow boundaries to change? (c_L)
Capital deepening (§3.1.5)	AI increases the capital used per worker through AI adoption and capital-intensive AI production.	1. How much does AI increase capital intensity? (s_K) 2. Does AI substitute for or complement labor? (s_K) 3. Who owns the capital and captures its returns? (c_K)

Stage 2: Who Captures the Initial Productivity Gains?

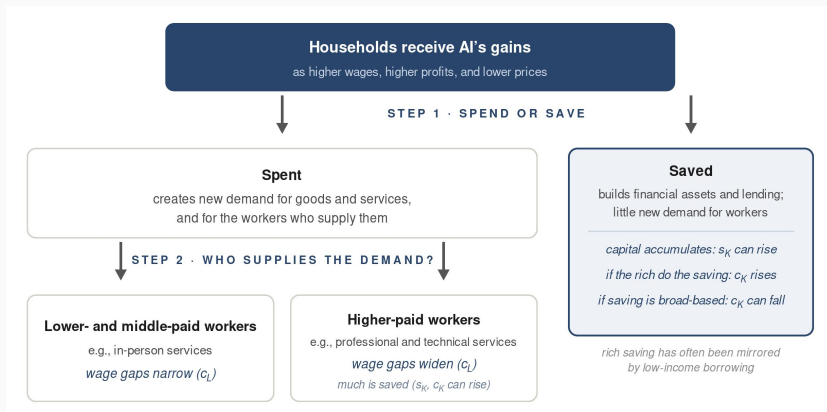
- AI suppliers, adopting firms, workers, and consumers (and within each group).
- Suppliers capture part of the gains upstream; the rest is split at adopting firms by labor-market power, product-market competition, and demand responsiveness.



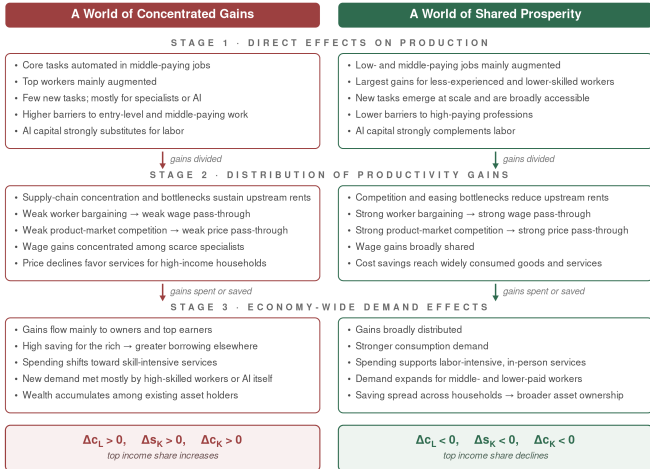
Stage 3: New Demand and Where It Lands

The gains re-enter the economy:

- Spending reshapes demand for goods and services, and for the workers who provide them.
- Savings can finance investment. But when concentrated among rich households, they may instead finance borrowing by non-rich households and the government (Mian et al., 2021).



Synthesis: Two Polar Scenarios



- Not forecasts: actual outcome will combine elements of both
- Illustrate how AI can produce very different distributional outcomes
- Asymmetry: concentrated gains require continuation of recent trends; shared prosperity requires a reversal

Empirical Indicators Across the Framework

1 Direct effects on production

Automation vs. augmentation (AI usage) · breadth of adoption (surveys) · wages, employment · entry-level hiring · new tasks · skill requirements

2 Division of the gains

Supplier margins & bottlenecks · wages vs. productivity in adopters · prices & markups · demand response in exposed sectors

3 Economy-wide demand

Household spending, saving & debt · composition of spending · where employment grows

The scoreboard

The three channels (c_L , s_K , c_K) in distributional national accounts.

Existing Datasets and Measurements (+ Remaining Gaps)

What already exists:

Official statistics

BLS: jobs, wages, prices
CPS & expenditure surveys
Census BTOS adoption
Distributional accounts

New ecosystem

Provider usage: Anthropic,
OpenAI, Google
Postings: Lightcast, Revelio
Compute: Epoch AI
Payroll: ADP

Early integration

Stanford AI Indicators
Canaries dashboard
Linked postings & resumes

Remaining gaps:

Some Poorly Measured, Others Poorly Linked

Poorly measured: productivity gains, automation costs, new tasks, supply-chain rents, demand responsiveness, and AI-capital ownership.

Poorly linked: few datasets connect AI use and labor-market outcomes to market power, bottlenecks, ownership, and sectoral income and price elasticities.

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